

Sec 3.7 – Implicit Functions

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

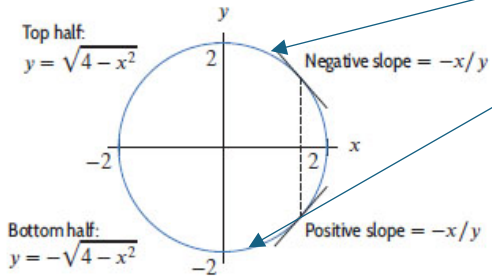
Up until now, the functions we've worked with have been of the form $y = f(x)$. We say that y is an **explicit** function of x .

In this section, we will consider equations such as $x^2 + y^2 = 4$. This gives y and an **implicit** function of x .

Find y when $x=1$.

$$1^2 + y^2 = 4 \quad y^2 = 4 - 1 \quad y^2 = 3 \quad y = \pm\sqrt{3}$$

In the graph, there are x values which correspond to two y values, and so y is not a function of x on the whole circle.



If we solve for y , we get: $y = \pm\sqrt{4 - x^2}$, where $y = +\sqrt{4 - x^2}$ is the top half of the circle and $y = -\sqrt{4 - x^2}$ is the bottom half of the circle.

y is one function on the top half, and a different function on the bottom half.

The equation does represent a curve which has a tangent line at each point. The slope of the tangent line can be found by differentiating the equation of the circle with respect to x . This process is called **implicit differentiation**.

$$\frac{d}{dx}x^2 + \frac{d}{dx}y^2 = \frac{d}{dx}4$$

We can think of y as a function of x and use the chain rule, where $y(x)$ is the “inner” function, $2x + 2y \frac{dy}{dx} = 0$

We need to solve for $\frac{dy}{dx}$ $2y \frac{dy}{dx} = -2x$, $\frac{dy}{dx} = -\frac{x}{y}$. This means the derivative at a point depends on both x and y . The derivative of this function exists everywhere that $y \neq 0$, and not at the points $(2,0)$ and $(-2,0)$, where there are vertical tangent lines.

Now, using the work above, find the equation of the tangent line to $x^2 + y^2 = 4$ when $x=1$ and $y=\sqrt{3}$.

We have $\frac{dy}{dx} = -\frac{x}{y}$, and to determine the slope we plug in the x and y values.

$$\left. \frac{dy}{dx} \right|_{(1, \sqrt{3})} = -\frac{1}{\sqrt{3}}$$

We have the point, $(1, \sqrt{3})$ and the slope at that point, $-\frac{1}{\sqrt{3}}$, so using the point-slope form for the equation of a line gives us:

$$y = m(x - a) + b$$

$$y = -\frac{1}{\sqrt{3}}(x - 1) + \sqrt{3}$$

Partner up

Find $\frac{dy}{dx}$ for each function:

1. $x^3 + y^3 = 8$

2. $y = x^2y^3 + x^3y^2$ *Be careful: which derivative rules do you need?*

3. $(x - y)^2 = x + y - 1$

4. $\frac{x - y^3}{y + x^2} = x + 2$

You don't have to use the quotient rule! Use algebra first!

5. $y = \sin(xy)$

6. $x = 3 + \sqrt{x^2 + y^2}$

1. $x^3 + y^3 = 8$

Solution: Take the derivative of both sides and use the chain rule:

$$3x^2 + 3y^2 \frac{dy}{dx} = 0.$$

Solve for $\frac{dy}{dx}$:

$$3y^2 \frac{dy}{dx} = -3x^2$$

$$\frac{dy}{dx} = \frac{-3x^2}{3y^2} = -\frac{x^2}{y^2}$$

2. $y = x^2y^3 + x^3y^2$

Solution: Take the derivative of both sides - this example requires the product rule.

$$\frac{dy}{dx} = (x^2)'y^3 + x^2(y^3)' + (x^3)'y^2 + x^3(y^2)'$$

$$\frac{dy}{dx} = 2xy^3 + x^2(3y^2)\frac{dy}{dx} + 3x^2y^2 + x^3(2y)\frac{dy}{dx}$$

Solve for $\frac{dy}{dx}$:

$$\frac{dy}{dx} - x^2(3y^2)\frac{dy}{dx} - x^3(2y)\frac{dy}{dx} = 2xy^3 + 3x^2y^2$$

$$(1 - 3x^2y^2 - 2x^3y)\frac{dy}{dx} = 2xy^3 + 3x^2y^2$$

$$\frac{dy}{dx} = \frac{2xy^3 + 3x^2y^2}{1 - 3x^2y^2 - 2x^3y}$$

3. $(x - y)^2 = x + y - 1$

Solution: Take the derivative of both sides:

$$2(x - y)(x - y)' = 1 + \frac{dy}{dx} \Rightarrow 2(x - y)\left(1 - \frac{dy}{dx}\right) = 1 + \frac{dy}{dx}$$

Solve for $\frac{dy}{dx}$:

$$2(x - y) - 2(x - y)\frac{dy}{dx} = 1 + \frac{dy}{dx}$$

$$-2(x - y)\frac{dy}{dx} - \frac{dy}{dx} = 1 - 2(x - y)$$

$$(-2(x - y) - 1)\frac{dy}{dx} = 1 - 2(x - y)$$

$$\frac{dy}{dx} = \frac{1 - 2(x - y)}{-2(x - y) - 1}$$

4. $\frac{x - y^3}{y + x^2} = x + 2$ (You don't have to use the quotient rule! Use algebra first!)

First we multiply by the denominator to clear fractions:

$$x - y^3 = (x + 2)(y + x^2) \text{ or } x - y^3 = xy + 2y + x^3 + 2x^2.$$

Now we take derivatives of both sides:

$$1 - 3y^2 \frac{dy}{dx} = (x)'y + x(y)' + 2 \frac{dy}{dx} + 3x^2 + 4x \text{ or}$$

$$1 - 3y^2 \frac{dy}{dx} = y + x \frac{dy}{dx} + 2 \frac{dy}{dx} + 3x^2 + 4x$$

Solve for $\frac{dy}{dx}$:

$$-3y^2 \frac{dy}{dx} - x \frac{dy}{dx} - 2 \frac{dy}{dx} = y + 3x^2 + 4x - 1$$

$$(-3y^2 - x - 2) \frac{dy}{dx} = y + 3x^2 + 4x - 1$$

$$\frac{dy}{dx} = \frac{y + 3x^2 + 4x - 1}{-3y^2 - x - 2}$$

5. $y = \sin(xy)$

Take the derivative of both sides:

$$\frac{dy}{dx} = \cos(xy)(xy)' \text{ or } \frac{dy}{dx} = \cos(xy)((x)'y + x(y)') = \cos(xy)(y + x \frac{dy}{dx})$$

Solve for $\frac{dy}{dx}$:

$$\text{Distribute: } \frac{dy}{dx} = \cos(xy)y + \cos(xy)x \frac{dy}{dx}$$

$$\text{Move } \frac{dy}{dx} \text{ to one side: } \frac{dy}{dx} - \cos(xy)x \frac{dy}{dx} = \cos(xy)y$$

$$\text{Factor: } \frac{dy}{dx}(1 - x \cos(xy)) = y \cos(xy)$$

$$\text{Divide: } \frac{dy}{dx} = \frac{y \cos(xy)}{1 - x \cos(xy)}$$

6. $x = 3 + \sqrt{x^2 + y^2}$

Solution: Take the derivative of both sides and use the chain rule:

$$1 = \frac{1}{2}(x^2 + y^2)^{-1/2}(x^2 + y^2)', \text{ which simplifies to}$$

$$1 = \frac{1}{2}(x^2 + y^2)^{-1/2} \left(2x + 2y \frac{dy}{dx} \right).$$

Solve for $\frac{dy}{dx}$:

$$\frac{1}{2}(x^2 + y^2)^{-1/2} 2x + \frac{1}{2}(x^2 + y^2)^{-1/2} 2y \frac{dy}{dx} = 1$$

$$(x^2 + y^2)^{-1/2} y \frac{dy}{dx} = 1 - (x^2 + y^2)^{-1/2} x$$

$$\frac{dy}{dx} = \frac{1 - (x^2 + y^2)^{-1/2} x}{(x^2 + y^2)^{-1/2} y}$$

Alternate solving for $\frac{dy}{dx}$:

$$\frac{1}{\frac{1}{2}(x^2 + y^2)^{-1/2}} = 2x + 2y \frac{dy}{dx}$$

$$\frac{1}{\frac{1}{2}(x^2 + y^2)^{-1/2}} - 2x = 2y \frac{dy}{dx}$$

$$\frac{\frac{1}{\frac{1}{2}(x^2 + y^2)^{-1/2}} - 2x}{2y} = \frac{dy}{dx}$$